Algorithmic Trading Research and Execution

Causal strategy research across five instruments, shared capital and synthetic execution

Kartik Vij | 30 October 2026

This C++20 study evaluates fixed intraday momentum, continuation and opening-anchor reversion rules, with a separate daily-risk allocation experiment. A registered five-instrument archive spans 2016-2025: 2016 initializes histories, 2017-2023 supplies development observations, 2024 is validation and 2025 is a frozen final retrospective evaluation. C++ owns signals, accounts, matching and durable recovery; Python independently verifies arithmetic and formats recorded evidence. The final evaluation retains 77 primary and 345 secondary/matched series, fourteen variants, 200 risk-account and 50 shared-capital cases. Some index point means are positive, but concentration, unavailable matched outcomes and controlled bootstrap size excess prevent a supported advancement claim. Separate synthetic process-broker and execution studies establish tested accounting/recovery invariants, not observed broker fills. Archived Release benchmarks measure bounded local workloads. Financial results, source coverage, numerical uncertainty and software correctness remain separate conclusions.

1. Research question and contribution

2. Data, information sets and account conventions

3. Trend rules, transaction costs and development findings

4. Reversion model and finite-sample identification

5. Registered reversion decision and historical comparison

6. Lagged risk, corporate actions and allocation value

7. Statistical interpretation and consequential assumptions

8. Multi-instrument development and frozen evaluation

9. Exact accounting, broker protocol and durable recovery

10. Synthetic execution measurements and censoring

11. Implementation performance and resource limits

12. Reproduction, verification and evidence access

13. Findings, limitations and completion boundary

References

1. Research question and contribution

The research question is whether fixed, prior-only intraday methods provide an economically material improvement over explicit controls after declared costs, delays and exposure conventions. A separate question is whether their targets can be translated into an exact order/account process with deterministic recovery. The source and report distinguish hypothetical historical references, synthetic execution and measured local processing costs.

The original registered SPY study uses inspected 2022-2023 observations and is retained as a reference. The extension fixes SPY, QQQ, IWM, TLT and GLD before replay. Three equity ETFs are correlated, and prior GLD research means the later panel is not globally unseen. The frozen sequence prevents selection using this extension's final outcomes; it cannot undo prior research exposure.

The contribution is an inspectable chain from validated observations to causal decisions, funded ledgers, matched controls, uncertainty and independently reconstructed recovery. A failed economic screen or unavailable account tail remains a result, rather than being replaced by another cost, instrument or parameter setting.

2. Data, information sets and account conventions

This chapter describes the original SPY reference conventions. The separate five-instrument archive contains 2,514 exchange sessions and retained raw acquisition/action evidence for 2016-2025. The multi-instrument chapter specifies its operational masks, canonical corporate actions, source amendments and later-period scope. Original SPY results retain their own snapshot and prior-use exclusions.

Historical inputs are Alpaca SIP SPY one-minute raw-price bars in USD. A bar timestamp is its interval start; its close, range and volume become available only when that minute completes. Exchange calendars determine each session and its New York daylight-saving conversion. Historical final bars are not a record of original live receipt and revision times.

The 2022-2023 calendar contains 501 sessions. Intraday studies retain 496 complete 390-minute sessions, totaling 193,440 bars. Three early closes, the previously inspected 2023-01-03 diagnostic and 2023-06-05 are excluded. The latter has four persistently missing minute starts after repeat and bounded gap probes; its whole session is omitted under a declared policy revision before aggregate scoring. Missingness is not repaired with prices, another feed or a fabricated cash day.

Momentum scores all 496 selected dates. Continuation and reversion use twenty eligible sessions only for initial history and score 476 dates. Their histories include strictly earlier eligible completed sessions. Risk allocation instead preserves all 501 exchange dates, with 252 calibration closes and a twenty-session warm-up. Twenty risk dates use the registered daily-only fallback when native RV5 history is incomplete; valid dated account outcomes are retained.

Convention	Intraday research	Daily risk	Execution
Quantity/account	Continuous notional; reinvested fractional equity	Fractional shares, cash and dividend receivables	Whole shares; exact integer money
Price/fill	Delayed minute-open trade-price proxy	Raw references plus corporate-action accounting	Constructed top-of-book quotes and incremental fills
Cost	0, 0.5, 1, 2, 5 bp per side; 1 bp primary	Same fixed cost grid applied to traded notional	Explicit per-fill fees; zero in synthetic paper scenarios
Holding scope	Flat intraday; cash earns zero	Overnight holdings; zero cash interest	Single symbol, unlevered long inventory
Period role	Inspected 2022-2023 development	2021 calibration, 2022-2023 development	Synthetic observations only

Local 2024-2025 strategy prices and candidate performance remain unopened. Early-2024 corporate-action processing metadata only reconciles entitlements earned before the development endpoint. Earlier pilot inspection, previous designs and literature exposure are recorded; registration does not turn this development period into an untouched test.

3. Trend rules, transaction costs and development findings

SPY-IM-01 observes the open-to-15:30 move, enters long only when it is positive, and liquidates at the 15:55 decision. Primary delayed references are 15:31 and 15:56 opens; the fixed two-minute stress uses 15:32 and 15:57. Its predictor differs from the previous-close-based early-return measures in the motivating literature [1,2].

$$m_d = C_{d,\,15:29}/O_{d,\,09:30} - 1, \qquad z_d = \mathbf{1}_{\{m_d > 0\}}$$

Only the completed 15:29 bar close enters the 15:30 decision.

SPY-TOD-01 adapts the same-clock scale/activity idea recorded in its motivating source [10]; it does not replicate that paper's full strategy. It evaluates every thirty minutes from 10:00 to 15:30. It compares the current open-to-clock move with the mean absolute same-clock move from twenty prior sessions. The primary volume variant also requires cumulative volume above the same-clock historical median for fresh entries. Volume does not determine exits. Both variants liquidate at 15:55; re-entry is allowed at later decision clocks. Unsigned volume measures activity, not signed order flow or option-dealer inventory.

$$u_{d,m} = C_{d,m}/O_d - 1, \quad s_{d,m} = \frac{1}{20}\sum_{j=d-20}^{d-1} |u_{j,m}|$$

The price condition is u > s. The volume condition is current cumulative shares divided by the prior same-clock median greater than one.

$$R(c) = \frac{P_x(1-c)}{P_e(1+c)} - 1, \qquad c = k/10^4$$

Each closed trade pays both sides. Within-session returns compound; genuine cash dates return zero. Costs are hypothetical all-in assumptions.

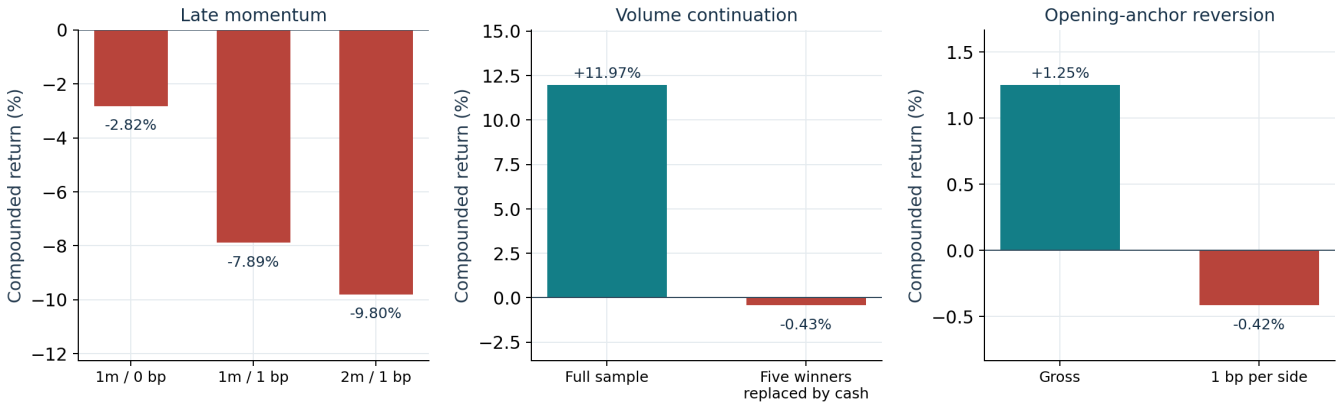


Figure 1. Recorded development summaries in different trading windows, not a risk-matched strategy ranking. Momentum uses 496 dates; continuation and reversion use 476. The continuation counterfactual replaces its five largest winning dates with cash.

Momentum loses 2.8207% even at zero cost and 7.8924% at the primary cost/delay. Volume continuation has a positive 11.965217% primary point return, but both variants fail their registered directional/robustness screens. Its net advantage over price-only is 1.493531 bp per scored day: 0.758595 bp gross improvement and 0.734936 bp reduced cost drag. Paired bootstrap intervals include zero. The positive tail and lower turnover explain part of the result; they do not establish volume alpha.

4. Reversion model and finite-sample identification

SPY-MR-01 fixes an opening anchor to the volume-weighted mean of the first thirty minute closes. This is a bar-close proxy, not transaction VWAP. Five-minute residuals run from the 10:00 completed close through noon. The current noon residual is evaluated using an unrestricted pooled AR(1) fitted to twenty strictly earlier sessions: 24 transitions per session, 480 pairs, and no cross-session pair.

$$A_d = \frac{\sum_{i=0}^{29} V_{d,i} C_{d,i}}{\sum_{i=0}^{29} V_{d,i}}, \qquad X_{d,k} = \log C_{d,29+5k} - \log A_d$$

$$X_{k+1} = a + \phi X_k + \varepsilon_{k+1}, \qquad \widehat{v}_\varepsilon = \mathrm{SSE}/(N-2)$$

Centered OLS retains the raw intercept, slope and variance. Singular designs, nonpositive slopes, a unit-root boundary, slopes above one and zero innovation variance remain distinct failures. Positive innovation variance and a slope strictly between zero and one permit an OU interpretation, with the registered floating-point tolerances. This numerical admissibility is not a stationarity test.

$$dX_t = \kappa(\theta - X_t)\,dt + \sigma\,dW_t, \qquad \phi = e^{-5\kappa}$$

The continuous OU conditional mean implies the discrete five-minute autoregressive mapping. Diffusion volatility sigma is distinct from discrete innovation variance v_epsilon.

$$\theta = \frac{a}{1-\phi}, \quad \kappa = -\frac{\log \phi}{5}, \quad t_{1/2} = \frac{\log 2}{\kappa}$$

$$g(x;\delta,H) = (\theta - x)e^{-\kappa\delta}(1 - e^{-\kappa H})$$

Subtracting conditional means at delay and delay plus holding horizon gives this expected log move. Applying the five-minute close fit to minute-open outcomes remains a model assumption.

Independent observation noise can attenuate an autoregressive slope and create apparent reversion. Common opening-anchor noise creates a different dependence pattern. The synthetic validation varies clean persistence, history length, opening transients, observation/anchor noise and dependent sessions. Each case has 1,000 point trials and 200 interval trials; nominal 95% coverage is not established.

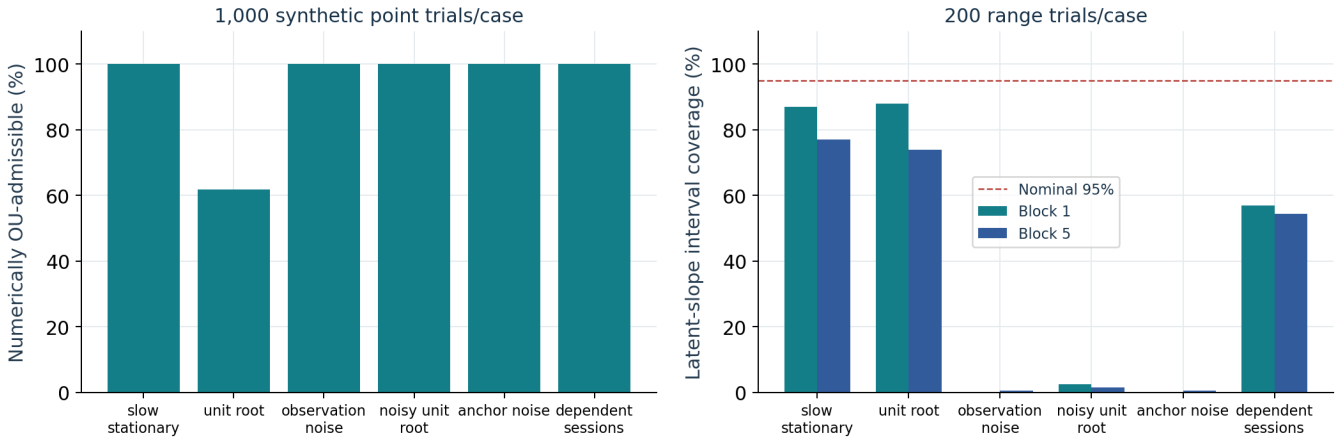


Figure 2. Selected pre-scoring synthetic controls. A latent unit root is numerically OU-admissible in 61.8% of clean trials and 100% with observation noise. Slope intervals substantially under-cover, especially under measurement misspecification.

5. Registered reversion decision and historical comparison

At noon, the candidate enters only if its prior fit is admissible, the current residual is below equilibrium and g(x;1,30) exceeds log(1.0001)-log(0.9999), the fixed primary two-sided log cost. The entry reference is 12:01 and the exit reference 12:31. Sign-only contrarian, no-reversion drift, always-long and cash controls retain their own required inputs. Unsupported fits produce a dated cash decision rather than disappearing from the result.

One action per policy/date is frozen before replaying ten cost/delay cases. The zero-cost and stress cases recost the same entries; they do not refit a cost-aware decision to each cheaper assumption. This distinguishes transaction-cost sensitivity from changing the strategy.

Policy	Active dates	Mean net bp/date	Compounded net	Daily-close drawdown
candidate	83	-0.0841	-0.4171%	2.8355%
sign-only	231	-0.6088	-2.9188%	5.8808%
drift	120	-0.3897	-1.8582%	2.9489%
always long	476	-2.7837	-12.5139%	14.2807%
Cash	0	0	0%	0%

The candidate is active on 83 of 476 dates, below the registered 100-date screen. There are 262 numerically admissible prior fits and 214 slopes above one. Candidate gross compounding is +1.2498%, while primary net compounding is -0.4171%. The positive-mean break-even cost is approximately 0.7588 bp per side; it is a sensitivity boundary, not an observed attainable execution cost. Annual primary net returns are -1.4441% in 2022 and +1.0421% in 2023.

Whole-date paired moving-block resampling uses 10,000 common-index replicates at lengths five and ten. Both ranges include zero for candidate mean net return and its net increments over sign-only and drift. The candidate gross increment over sign-only is negative. On the 262-date common forecast mask, candidate squared and absolute loss point estimates are worse than drift, but corresponding intervals include zero; statistically conclusive inferiority is not established either.

The registered assessment is cost_dominated and does not advance. The 100-active-date threshold is a project coverage convention, not a theorem about sample adequacy. Declared stresses, annual consistency, uncertainty and concentration screens also remain visible when an earlier screen fails. The failure completes this comparison without licensing a new search on unopened periods.

6. Lagged risk, corporate actions and allocation value

SPY-RISK-01 separates forecast accuracy from allocation value. A daily measurement starts with one share at the prior regular-session close. A dividend D going ex adds to the session reference value, so the daily factor is (close + D)/previous close. RV5 partitions that reference through the overnight step, session open and five-minute endpoints to the actual close. Unpaid account entitlements remain receivables until payment; they are not spendable cash.

$$\mathrm{RV5}_d = \sum_i (\log V_{d,i} - \log V_{d,i-1})^2, \quad \mathrm{daily2}_d = \left(\log \frac{C_d + D_d}{C_{d-1}}\right)^2$$

$$h_d = \frac{1}{20}\sum_{j=1}^{20} v_{d-j}, \qquad w_d = \min\left(1, \frac{0.10}{\sqrt{252 h_d}}\right)$$

A target uses exactly twenty preceding exchange sessions, with a cutoff five minutes after the previous actual close. The native intraday estimate requires complete RV5 histories. Its fixed fallback is the daily-only estimate on twenty affected development dates; it never deletes the current account return. Static risk budget uses full-2021 daily2 once before development. Passive equity is a separate control.

$$\mathrm{CER} = 252\left(\bar{r} - \frac{3}{2}s_r^2\right), \qquad L(h, y) = y/h + \log h$$

CER is a registered mean-variance approximation, not a fitted investor utility. The displayed QLIKE form gives equivalent forecast rankings after dropping outcome-only constants.

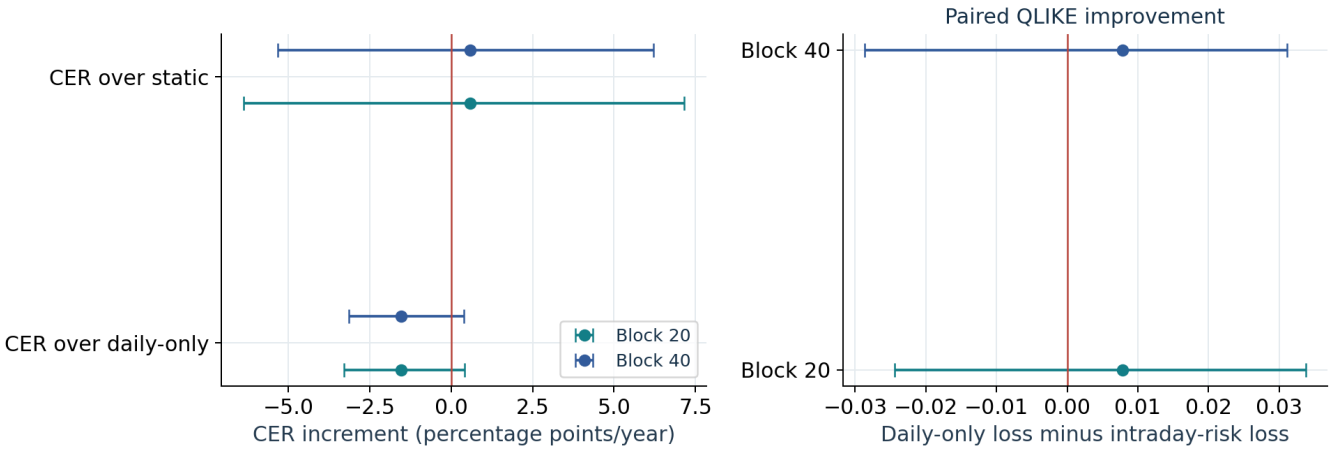


Figure 3. Registered paired 95% moving-block percentile ranges, 10,000 replicates at blocks 20 and 40. Point QLIKE improvement is positive, but neither forecast-loss nor economic-increment lower bounds establish advancement.

All forty account cases retain all 501 exchange dates; 479 dates support the primary native forecast comparison. At one-minute delay and 1 bp per side, intraday risk returns 1.7072%, daily-only 4.7624%, static risk 2.7920% and passive equity 2.3777%. These account returns are not directly comparable to the intraday strategy windows. Intraday risk CER is 1.5430 percentage points below daily-only and 0.5723 points above static risk. The primary QLIKE point increment is +0.007813, but both block intervals include zero.

The allocation study does not advance under revision 1. Different loss and utility metrics can rank the same rule differently. No cheaper cost case, recalibrated target or alternative realized-volatility partition is selected to rescue the registered result.

7. Statistical interpretation and consequential assumptions

Moving-block resampling retains complete dated records and common indices across candidates and controls [4]. It preserves dependence within sampled blocks, not the entire original dependence structure. The intraday experiments use lengths five and ten; risk allocation uses twenty and forty. Dates, genuine cash observations, control outcomes and forecast availability masks remain paired. Undefined contrast replicates invalidate a range rather than being discarded.

The studies use 10,000 outcome replicates and registered percentile conventions. Reversion parameter diagnostics additionally resample complete session sequences at blocks one and five, with 499 refits per historical range. Their synthetic undercoverage prevents treating them as calibrated latent-parameter confidence statements. Outcome intervals condition on the selected design and do not correct for the complete sequence of prior research trials.

Assumption	Why it matters	Interpretation retained
One ETF; two development years	Limited regimes, cross-sectional breadth and independent opportunities	No market-wide or out-of-sample alpha claim
Minute-open delayed proxies	No executable spread, depth, queue or original data-arrival record	No live execution inference from historical bars
0-5 bp hypothetical all-in costs	Can reverse economic signs; may be optimistic or conservative	Report fixed sensitivities without selecting their winner
Long/cash and zero cash interest	Changes attainable strategies, exposure and opportunity cost	Controls share declared conventions but need not be risk matched
100-date, annual and tail screens	Design/coverage judgments affect advancement	Keep raw findings and sampling precision beside the binary screen
Finite-sample AR/OU mapping	Noise and near-unit roots distort dynamics and uncertainty	Numerical admissibility is separate from economic predictability

Positive gross evidence, net value after assumed costs, interval precision and operational accounting each answer different questions. Passing software tests certifies stated contracts on tested inputs; failing a trading screen does not imply an implementation defect. Conversely, a favorable backtest point estimate cannot validate the mathematical observation model.

8. Multi-instrument development and frozen evaluation

The extension retains five instruments and three directional strategy families. Each model uses completed information and strictly earlier estimation histories. Fresh validation/final capital is independent of warmed features. The daily-risk experiment remains separate from the fifteen symbol/family owners in the whole-share shared account.

The original complete-minute intersection qualifies only four development years under a five-year gate. A separately registered amendment uses 33 required operational execution slots and reports feature availability separately. The parent assessment remains failed under its own definition. Three missing closes on 2019-08-12 remain unavailable after empty recovery responses; daily history resets instead of bridging those returns. Canonical issuer cash events, secondary split cross-checks and GLD cash-absence evidence retain their completeness assumptions.

Period	Role	Primary evaluation dates
2016	Initialize past-only histories	Not scored
2017-2023	Retrospective development	1,760
2024	Registered validation	252
2025	Frozen final retrospective evaluation	250

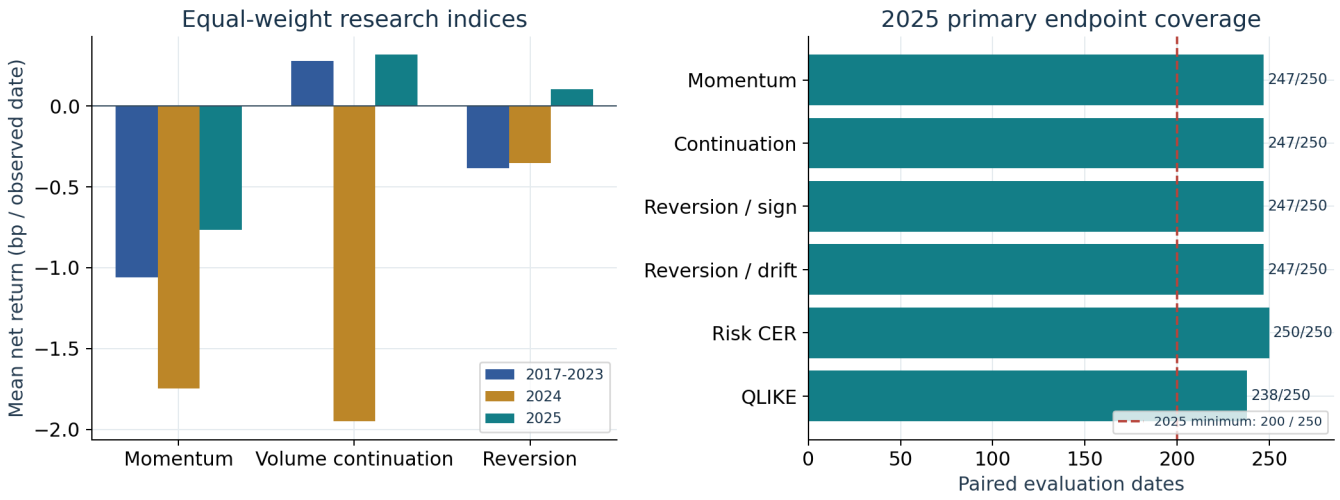


Figure 4. Five-instrument research indices at one basis point per side and one-minute delayed minute-open references. Mean net returns use each retained common-date mask; they are not shared funded wealth. Right: 2025 primary paired coverage. All six exceed the registered minimum of 200 dates and 80% of 250 sessions.

The 2025 pooled net index means are -0.7684 bp/day for momentum, +0.3157 for volume-conditioned continuation and +0.1042 for reversion, with 247 observed common dates. Zeroing each positive index's largest winning date changes its mean to -0.7021 and -0.0961 bp/day respectively, keeping the denominator fixed. This concentration diagnostic neither refits the strategy nor simulates a replacement funded ledger.

Fourteen one-factor sensitivity cases vary histories, observation grids and holding periods without selecting a replacement. Forty-five unit accounts support fifty funded risk-matched accounts. The sizing window retains 252 prior session slots; missing observations remain missing. Candidate and control exposure scale toward the smaller of their past annual volatility and the fixed 10% target, without leverage.

$$\sigma_t^* = \min(0.10, \widehat{\sigma}_{c,t-1}, \widehat{\sigma}_{b,t-1}), \qquad w_{j,t} = \sigma_t^* / \widehat{\sigma}_{j,t-1}$$

Dependence-aware inference resamples a common full calendar for the six-endpoint family, retaining unavailable slots. The reported lower bound is estimate minus a joint max-T critical value times its HAC standard error. Blocks 5, 10, 20 and 40, seed 290026 and 10,000 resamples remain fixed. The controlled calibration runs 20,000 outer families and detects nominal size excess in 19/20 case/block cells. Numerical bounds therefore do not support a validated nominal 95% simultaneous-confidence interpretation.

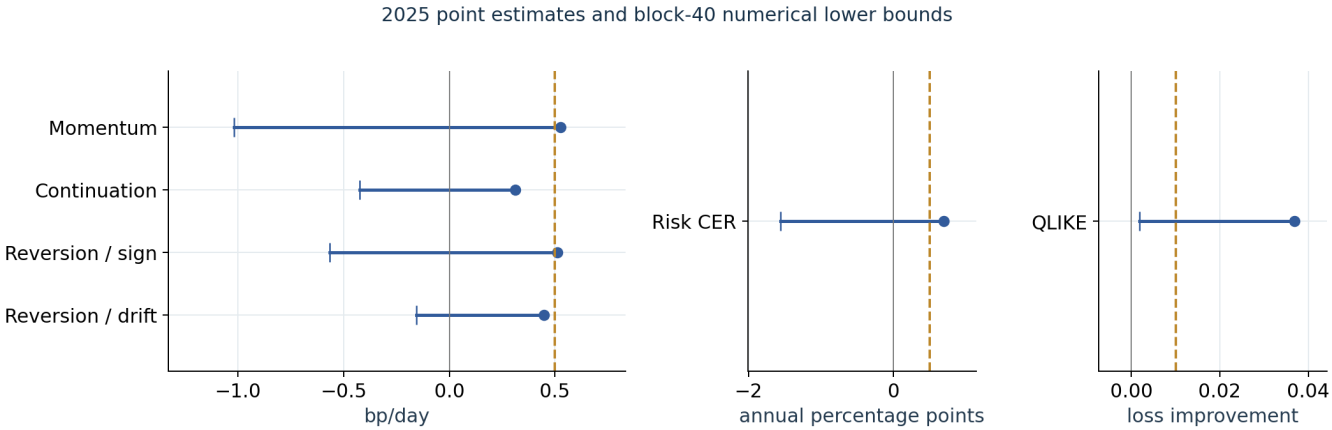


Figure 5. Final primary point estimates and block-40 numerical lower bounds. Dots are estimates, left ticks are lower bounds, and dashed gold lines are registered material increments. These are not two-sided intervals. The panels use distinct return, CER and QLIKE units. Size control failed; no endpoint lower clears its material increment.

Primary QLIKE coverage rises from 121/252 dates in 2024 to 238/250 in 2025. The final matched counts are 247, 247, 247, 0, 250 and 238 in registered endpoint order. The reversion-versus-drift endpoint is unavailable; the complete matched family remains unavailable rather than dropping that comparison. Positive relative point estimates do not establish incremental skill.

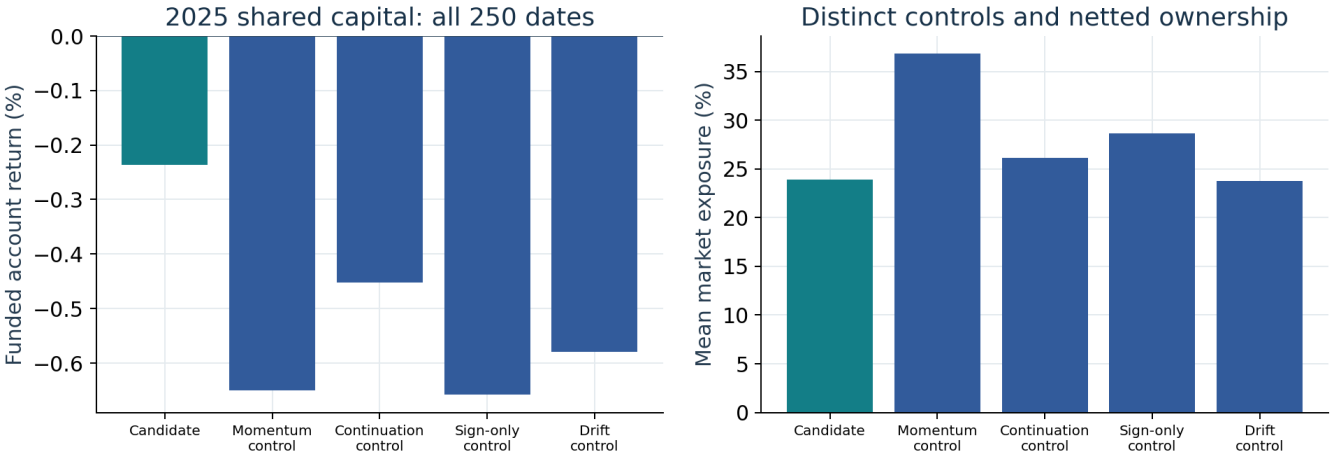


Figure 6. Five 2025 whole-share shared-capital accounts at the primary cost and delay, each starting from USD 100,000 and completing 250 dates. The right panel retains their different market exposure; return differences are not risk-matched alpha estimates.

All 200 risk-account and fifty shared-capital cases complete 2025. The primary shared candidate ends at USD 99,763.25481382, with USD 1,759.58798618 charged costs, 23.9159% mean exposure and 1.0196% drawdown. Earlier development paths and the primary 2024 shared path stop on missing required inputs; the 2024 primary completes only 79 dates. Complete 2025 paths do not fill those earlier tails.

The original M8 executable could not parse a legitimate 2026 payable date attached to a 2025 distribution. A corrected runner reproduced all 26 financial artifact identities and four 2024 scientific summaries before a separate replacement freeze authorized the final run. Original records remain retained. Only payment metadata crosses 2025; no 2026 price, session or forecast target is admitted.

9. Exact accounting, broker protocol and durable recovery

The five-instrument shared-capital book has one global cash balance, five inventories and fifteen strategy owners. Target netting and aggregate limits do not transfer ownership implicitly. A separate native child process owns the broker ledger and its durable history; the client recovers activity before trusting a snapshot. DAY, IOC and GTC limit policies, cancel/new replacement, fees and transport faults are tested through the shared matching core. Windows acceptance covers 313 checks in twelve families across 21 child processes. All events are synthetic; anonymous pipes demonstrate process separation, not external-broker connectivity.

The execution core owns one symbol, whole-share inventory, integer money, order and execution identities, reservations and a sequential trace. Legacy journals retain cents; the paper bridge uses USD 1e-9 units and an explicit zero-fee paper convention. Scope excludes shorts, leverage, fractional executions, corporate actions inside this execution account, exchange queue reconstruction and execution busts.

$$B_{\text{buy}} = q_{\text{remaining}}(P_{\text{limit}} + f_{\text{ceiling}}), \quad C' = C - sqP - F, \quad Q' = Q + sq$$

s = +1 for buys and -1 for sells. Fees F are nonnegative. Sell reservations cannot exceed held inventory; all intermediate integer operations are checked.

Input	State/resource rule
Submit / acknowledgement	Reserve first; pending_new becomes working. Late acknowledgements cannot regress later state.
Partial or full fill	Book each stable execution once, at valid limit/fee terms; reduce only the appropriate remaining reservation.
Cancel request / rejection	Retain reservations while pending; correlate a rejection to the current cancel identity.
Cancel confirmation	Release the remainder only when reported cumulative quantity agrees with received executions.
Contradiction / disconnect / restart	Latch the entry gate. Broker evidence remains processable; reconnect alone cannot reopen routing.
Exact reconciliation	Connected cash, position and complete open-order terms must agree; snapshots do not overwrite mismatches.

The protocol peer maintains independent broker state and finite FIFO top-of-book quote budgets. It generates API-shaped raw messages with stable client/broker/execution IDs. Scenarios cover fills, cancel races, duplicate delivery, uncertain submission resolved by ID lookup, missing-fill refusal, contradictory snapshots, deadlines and permanent stop. They are synthetic protocol tests, not observed external-broker order histories.

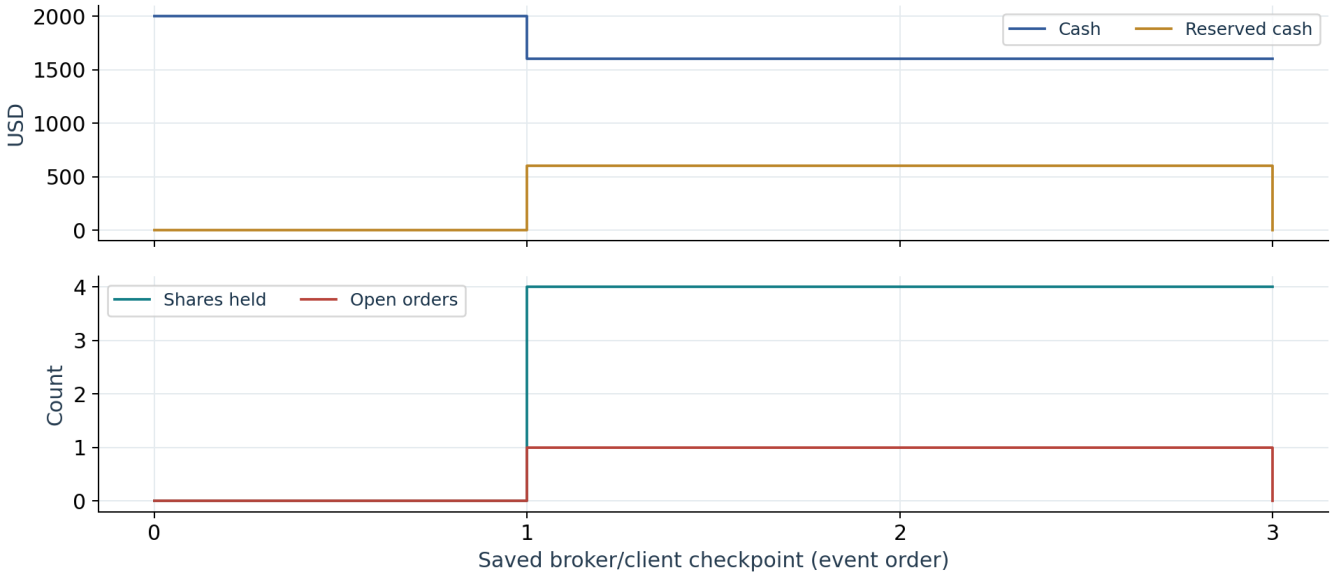


Figure 7. Saved checkpoints of the partial-fill/cancellation case. Cash and remaining reservations change for distinct reasons. The terminal account holds four shares with USD 1,600 cash after confirmation; it is not incorrectly flattened.

Checkpoint	Saved event	Cash / reserved USD	Held / open orders
0	Connected and reconciled	2000.00 / 0.00	0 / 0
1	Four shares filled; six remain	1600.00 / 600.06	4 / 1
2	Cancel request retains reserved cash	1600.00 / 600.06	4 / 1
3	Cancellation confirmed; four-share inventory remains	1600.00 / 0.00	4 / 0

A journaled owner evaluates an input on a candidate state, appends a sequence/hash-linked frame, synchronizes storage, then publishes state and returns dispatch eligibility. Recovery validates complete frames, replays through the same core and durably closes routing. An incomplete tail is recovered only by copying a verified prefix to another file; complete corruption is rejected. The original damaged evidence remains intact.

Once-only accounting requires serialized calls, stable nonconflicting execution IDs and replay under the same financial contract. External transmission can succeed before a response arrives, so local persistence cannot guarantee end-to-end exactly-once dispatch. Uncertain durable orders are not blindly resent. The interrupted and uninterrupted broker round trips finish flat at USD 2,000.24 from USD 2,000, an accounting oracle rather than a trading return.

10. Synthetic execution measurements and censoring

The extension separately evaluates 680 registered synthetic execution cases and 510 matched contrasts, with exact reconciliation of every terminal account. The historical cost/delay study retains 400 instrument/policy/cost/delay series and 240 paired contrasts. A fixed pre-2024 SIP quote sample retains 910 windows; 232 are eligible under the registered condition and timestamp rules. Sparse earlier-year eligibility is reported, not repaired by dropping the condition filter or fitting costs from final results. These studies have different evidence origins: sampled quote crossing references do not calibrate the synthetic fill/queue model. The following 144-case measurement is the original narrower reference.

The measurement grid contains 36 declared synthetic workloads: buy/sell directions, steady/favorable/adverse paths, and one-factor spread, displayed size, outbound/delivery or cancellation-processing changes. Four fixed policies combine crossing or same-side limits with five- or eleven-second cancellation deadlines. Matching reuses quote sequence budgets; advancing the clock does not replenish liquidity.

$$\mathrm{IS}_{\mathrm{executed}} = s\sum_i q_i(P_i - M_0) + F$$

$$\mathrm{IS}_{\mathrm{executed}} = s\sum_i q_i(P_i - M_i) + s\sum_i q_i(M_i - M_0) + F$$

$$\mathrm{IS}_{\mathrm{total}} = \mathrm{IS}_{\mathrm{executed}} + s\,q_{\mathrm{unfilled}}(M_T - M_0)$$

Arrival midpoint M0 is the benchmark; Mi is the fill-source quote midpoint; MT is the common terminal midpoint. Signed half-cent sums avoid midpoint rounding. Bps use requested notional, not filled notional.

Source and receipt timestamps are separate integer virtual seconds. First-fill and completion summaries show their available sample counts. Incomplete orders retain null completion times and an observation interval ending at confirmed cancellation or the horizon. Cancellation is policy-driven and informative; no independent-censoring survival estimator is fitted. Synthetic acknowledgement/fill delays are not wall-clock software latency.

Longer crossing deadlines increase aggregate filled shares from 72.22% to 76.67%, while both crossing policies have 3.5889 bp total marked shortfall in this grid. Passive coverage is 14.44%-20.00%, with substantial incomplete quantities. Opportunity marking uses a declared terminal price and is not realized P&L. Fees are zero. Impact, hidden liquidity, exchange priority and actual spreads or broker delays are not calibrated.

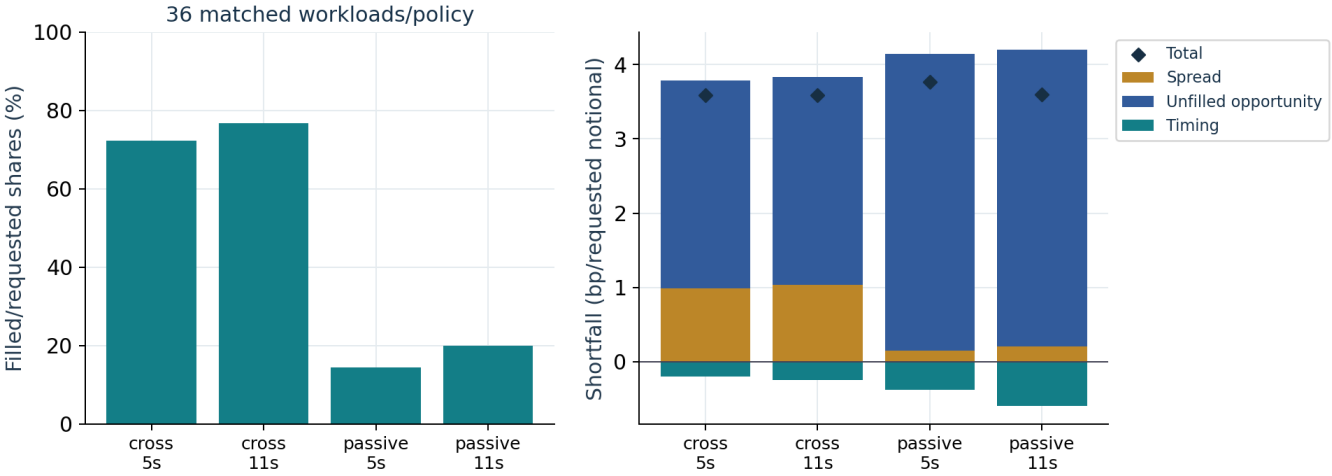


Figure 8. Equal-weight deterministic workload aggregates. Fill coverage and executed/unfilled costs are displayed together. These shares are constructed scenario frequencies, not empirical fill probabilities or a universal policy ranking.

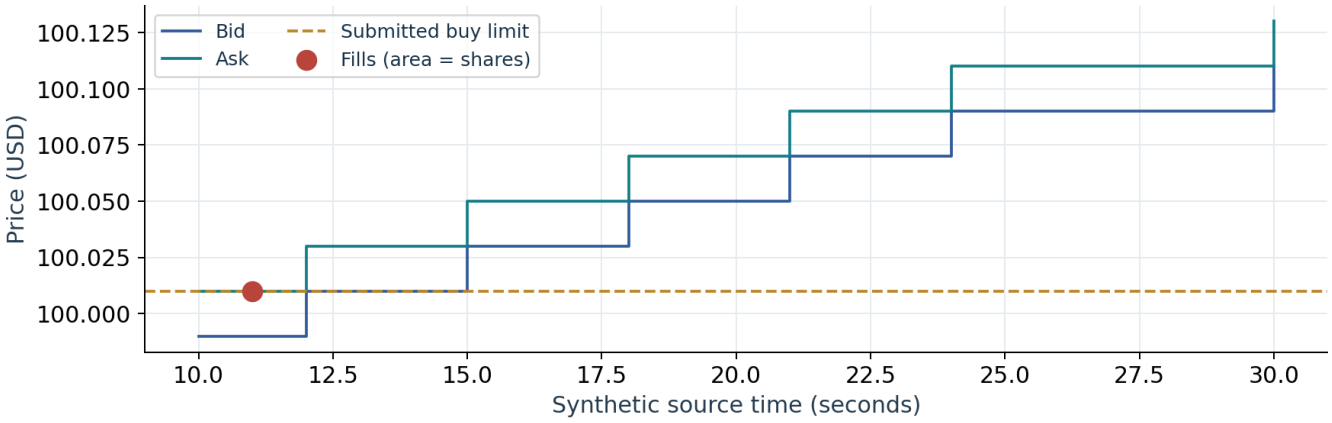


Figure 9. One adverse buy workload with crossing limit and five-second cancellation policy. Quote/fill source times are constructed. Fill marker area reflects shares. Quotes establish only the declared top-of-book budget.

Policy	Filled shares	Complete cases / 36	Total shortfall bp
cross_5s	72.22%	20	3.5889
cross_11s	76.67%	24	3.5889
passive_5s	14.44%	0	3.7667
passive_11s	20.00%	0	3.6000

11. Implementation performance and resource limits

Release workloads contain 8, 32 or 128 sequential round trips, five fresh-state repetitions and eight modes. Every round trip buys one share at USD 100, accepts and fills it, redelivers that fill, then sells one share at USD 100.01 with acknowledgement and fill. Typed paths have 7n+2 timed inputs; paper paths have 7n+3. Final cash must be USD 100,000 plus n cents, with zero inventory/reservations/open orders and 2n filled orders.

Mode family	Timed boundary
Core / extra candidate copy	Direct typed input, or additional outer runtime copy/apply/move. The core already copies internal state transactionally.
Volatile / synchronized journals	Candidate evaluation, serialization, SHA-256 framing and volatile append or real exclusive-writer append/flush. Both typed schema 2 and paper schema 3 are retained.
Warm restart	Open, read, validate, replay and synchronize a new disconnect barrier after an untimed file copy. Not a cold boot.

Steady-clock nanoseconds bracket each call. Nearest-rank p50/p95/p99 and raw samples are saved per trial; trial min/median/max describe variability. Timed-call throughput divides calls by summed call durations and excludes construction, between-call checks, final validation and export. A 1,000-sample empty-clock probe exposes timer scale. Instrumentation, warm caches, scheduling, antivirus and storage contribute to observed costs.

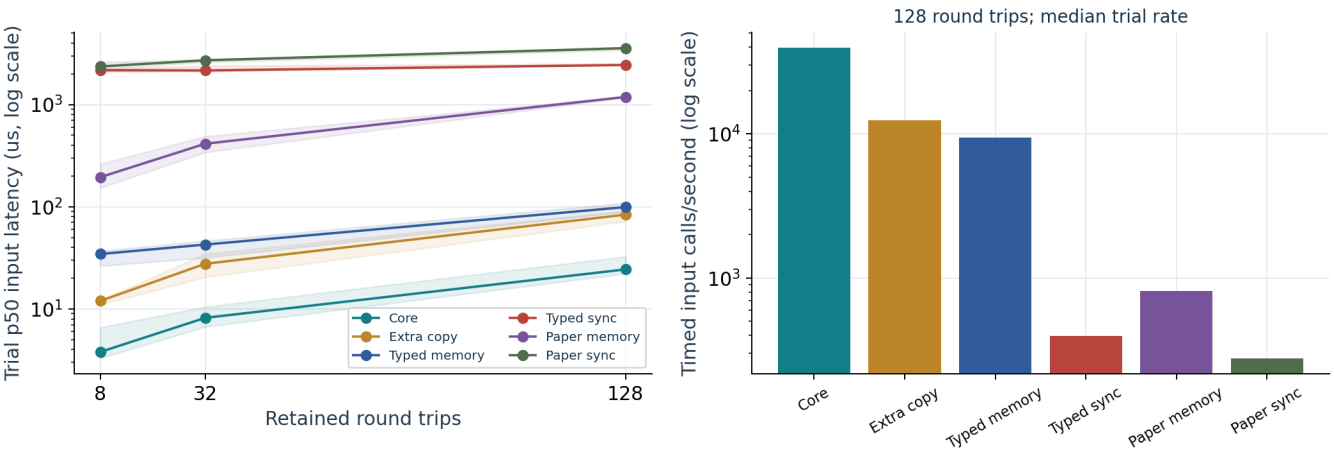


Figure 10. Native MSVC Release processing costs. Curves are medians of five trial p50 values with observed min/max bands; bands are not confidence intervals. Right: median per-trial timed-call rate at 128 round trips.

On the native Windows host, median trial p50 at 128 round trips is 24.4 us for direct core processing, 83.8 us with the additional copy, 2.4384 ms for synchronized typed journals and 3.5602 ms for synchronized paper journals. The report records /O2 /Ob2 /DNDEBUG, C++20 and warnings as errors. CPU metadata reports Intel Family 6 Model 183 Stepping 1 and 24 logical processors.

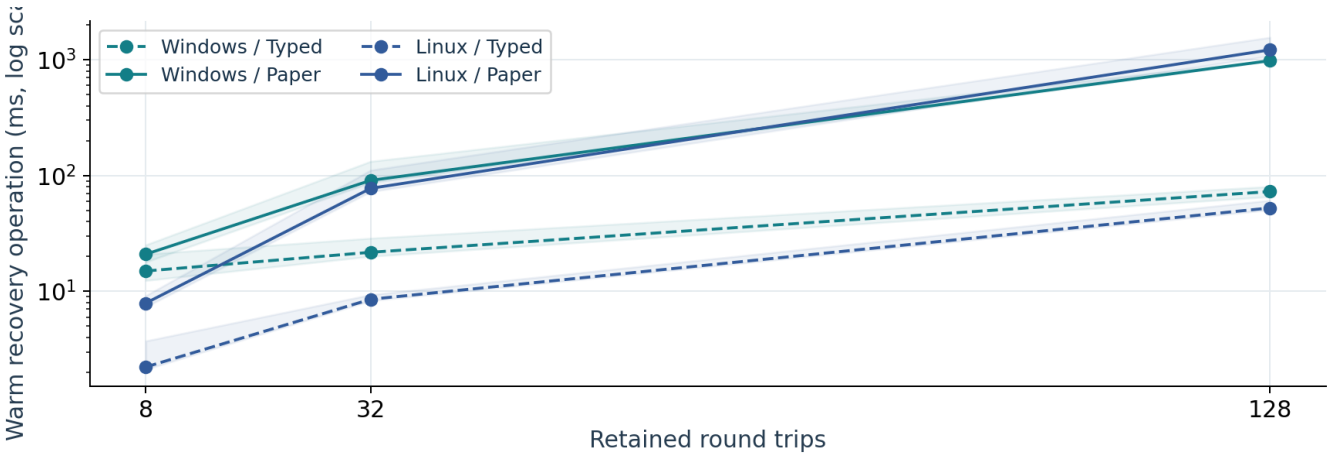


Figure 11. Warm replay costs on two different hosts/toolchains. Native paper recovery median is 982.58 ms; typed recovery 72.76 ms at 128 round trips. Cross-host differences cannot isolate an operating-system or compiler effect.

Mode at 128 round trips	Median trial p50, us	Trial p50 range, us	Median operations/s
core	24.4	22.2-32.6	39,536.83
copy_candidate	83.8	72.0-91.1	12,382.09
journal_memory	99.3	89.2-108.8	9,427.41
journal_sync	2,438.4	2,427.6-2,514.9	398.20
paper_memory	1,186.5	1,169.3-1,196.8	815.59
paper_sync	3,560.2	3,413.0-3,590.7	277.71
journal_recovery	72,763.5	65,784.9-80,413.4	13.74
paper_recovery	982,580.7	975,262.4-999,857.9	1.02

Native process-wide peak working set is 20.730 MiB; Linux observes 15.918 MiB in its reference run. These include sequential fixtures, results and allocator retention, not isolated per-case allocations or peaks. Journal caps remain 10,000 input records, 32 MiB per file and 64 KiB per frame; exhaustion stops append/restart. Rotation and compaction are absent. The tested grid does not establish throughput at those caps. No speed target or optimization is inferred from these measurements.

12. Reproduction, verification and evidence access

C++20 is authoritative under src/. Thin application entry points compose strategy, account, broker and report owners. Python performs build orchestration, independent numerical verification and publication transforms. One source list is discovered by build.py; Runtime.sln delegates to it. CMake and CTest are not used.

```
python -B build.py --configuration Release --target project_demo
scratch/native/Release/project_demo
python -B research/verify_project_demo.py <printed-directory>
python -B research/verify_publication.py
```

On Windows, install Desktop development with C++, a Windows SDK and Python 3.10+, open Runtime.sln, choose Release | x64 and build. Ctrl+F5 runs the default demo. The equivalent terminal commands use py -3 -B and executables have the .exe suffix. Linux requires a C++20 compiler, ar and OpenSSL development headers. Build warnings are errors. Debug and sanitized builds test correctness, not Release timing.

The first build downloads pinned nlohmann/json 3.12.0; --json-include supplies an existing validated include directory. Offline demos require no market credentials, broker service or private archive after dependencies are available. Historical reproduction additionally requires retained response pages, calendars, masks, corporate actions and registered manifests. Public summary tables cannot substitute for those raw inputs.

```
python -m pip install -r research/report_requirements.txt
python -B research/build_report.py --demo <verified-demo-directory>
python -B research/verify_publication.py
```

The publication reads one content model and a hash-checked evidence manifest. Figures convert recorded units and group retained results; they do not score strategies or run accounts. The self-contained HTML embeds saved synthetic event views. The PDF provides corresponding static figures. UTF-8/LF hashes bind distributed text; original inputs, run records and PDFs retain raw-byte identities.

Original and replacement validation/final directories remain local. Full local verification authenticates raw provider data, causal decisions and ledgers. Uploaded compact bundles support independent derived arithmetic and plot review; they omit sixteen detailed source artifacts per evaluation and the frozen Windows executable bytes. This distinction remains explicit in the evidence records. Frozen final reproduction uses its retained runner, not an ordinary rebuilt executable.

Runtime figures retain their original Windows and GNU/Linux host/toolchain identities. They benchmark the original single-symbol typed/paper calls and recovery workloads, not the expanded five-instrument process peer. The extension has no new speed or throughput claim. Detailed contracts, exact commands, input provenance and accepted hashes are linked from docs/running.md, docs/architecture.md and docs/validation.md.

13. Findings, limitations and completion boundary

The complete study distinguishes economic findings from engineering acceptance. The original SPY screens fail. In the five-instrument final evaluation, volume-conditioned continuation and reversion have small positive pooled index means, but both are concentrated in their largest winning date. The primary shared candidate loses 0.2367% after its declared charges. No final primary numerical lower bound exceeds its material threshold.

The final primary family has adequate coverage; the matched family does not. Controlled calibration detects size excess, so nominal simultaneous-confidence claims are withheld. Registered sensitivity cases and unavailable outcomes remain in the record. No new rule, cost, instrument subset or delay is selected after final inspection.

The runtime demonstrates whole-share accounting, explicit ownership and netting, an independently journaled synthetic child broker, bounded lifecycle policies, exact terminal reconciliation and deterministic tested recovery. Synthetic liquidity, anonymous-pipe transport and local timings do not establish real-market fills, external-broker compatibility or a high-frequency latency target.

Material limits include selected and correlated instruments, prior market/strategy exposure, historical final-bar information, missing references, hypothetical costs, fractional research versus whole-share execution, no cash yield, limited OU/noise identification, uncalibrated synthetic liquidity and bounded journals without rotation. The registered archive ends in 2025. The completed scope contains the research, risk matching, shared capital, broker simulation, cost/execution comparisons, controlled inference calibration, frozen retrospective evaluation and this consolidated publication.

References

[1] Gao, Han, Li and Zhou (2018). Market intraday momentum. Journal of Financial Economics 129, 394-414. Motivation; this project uses different opening and holding conventions.
https://doi.org/10.1016/j.jfineco.2018.05.009

[2] Baltussen, Da, Lammers and Martens (2021). Hedging demand and market intraday momentum. Journal of Financial Economics 142, 377-403. Mechanism motivation, not a replication.
https://academicweb.nd.edu/~zda/intramom.pdf

[3] Moreira and Muir (2017). Volatility-managed portfolios. Journal of Finance 72, 1611-1644. Motivation for separating risk allocation from return prediction.
https://www.nber.org/papers/w22208

[4] Kunsch (1989). The jackknife and the bootstrap for general stationary observations. Annals of Statistics 17, 1217-1241. Dependent-data resampling context.
https://doi.org/10.1214/aos/1176347265

[5] Frozen intraday momentum protocol revision 2, data policy and continuation protocol revision 1. Governing local definitions and prior-use declarations.
https://github.com/KVInator/algorithmic-trading-runtime/tree/main/research

[6] Frozen SPY-MR-01 and SPY-RISK-01 revision-1 protocols/settings. These records govern the exact implemented rules and failure screens.
https://github.com/KVInator/algorithmic-trading-runtime/tree/main/research

[7] Alpaca historical stock bars. Provider conventions; retained raw evidence remains required for historical reproduction.
https://docs.alpaca.markets/us/reference/stockbars

[8] Alpaca order and streaming specifications. The implemented local message subset is synthetic and does not verify the current external service.
https://docs.alpaca.markets/docs/working-with-orders

[9] Project source, operational contracts and curated evidence manifest. Exact source-document and native-output hashes accompany the publication.
https://github.com/KVInator/algorithmic-trading-runtime/tree/main/docs/evidence

[10] Zarattini, Aziz and Barbon. Beat the Market: An Effective Intraday Momentum Strategy for S&P500 ETF (SPY), version dated 2025-02-03 in the frozen continuation protocol. Method motivation; the implemented long/cash rules and gates remain those of SPY-TOD-01.
https://alexandria.unisg.ch/bitstreams/a99aba00-f967-49b3-aceb-f544dc386e0b/download